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Introduction & Background

The objective of this project is to design, minimize, and implement combinational logic circuits based on real-world control specifications. Digital logic design relies on the translation of system requirements into truth tables, followed by simplification using Karnaugh Maps (K-maps) to ensure efficiency and reduce gate count.

In this report, we present the design and implementation of two distinct control systems:

1. **Scenario 2: Access Control & Indicators:** A security system managing a lab door, badges, and emergency protocols.
2. **Scenario 3: Mini Vending Controller:** A control logic for a vending machine that handles coin validation, stock monitoring, and dispensing mechanisms.

Our goal for both scenarios is to derive the minimal Sum-of-Products (SOP) equations and verify the logic through CircuitVerse simulations.

Scenario 2: Access Control & Indicators

1. Scenario Description

The Access Control System utilizes four binary inputs and six binary outputs.

2. Inputs:

- A (Admin Badge): High (1) if an admin badge is detected.
- G (Guest Badge): High (1) if a guest badge is detected.
- D (Door State): High (1) if the door is physically open.
- F (Fire Alarm): High (1) if a fire is detected.

3. Outputs:

- Grant: Signals access granted.
- Deny: Signals access denied.
- RedLED: Status indicator (Error/Deny/Evacuation).
- GreenLED: Status indicator (Success).
- Lock: Controls the electromechanical lock (1 = Locked).
- Evacuate: Emergency signal.

4. System Priorities:

1. Emergency (F=1): System unlocks door, activates evacuation signal and RedLED, disables standard indicators.

2. Normal Operation (F=0):

- Valid badge (A or G): Grant access, unlock door, Green LED on.
- No badge: Deny access, lock door, Red LED on.

5. Truth Table Verification

A 16-row truth table was constructed to verify all possible combinations of inputs.

A	G	D	F	Grant	Deny	RedLED	GreenLED	Lock	Evacuate
0	0	0	0	0	1	1	0	1	0
0	0	1	0	0	1	1	0	1	0
0	1	0	0	1	0	0	1	0	0
0	1	1	0	1	0	0	1	0	0
1	0	0	0	1	0	0	1	0	0
1	0	1	0	1	0	0	1	0	0
1	1	0	0	1	0	0	1	0	0
1	1	1	0	1	0	0	1	0	0
0	0	0	1	0	0	1	0	0	1
0	0	1	1	0	0	1	0	0	1
0	1	0	1	0	0	1	0	0	1
0	1	1	1	0	0	1	0	0	1
1	0	0	1	0	0	1	0	0	1
1	0	1	1	0	0	1	0	0	1
1	1	0	1	0	0	1	0	0	1
1	1	1	1	0	0	1	0	0	1

6. K-Map Simplification & SOP Equations

- Grant

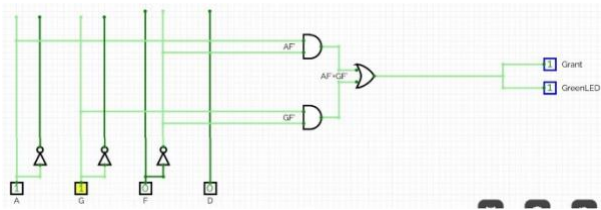
0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

- Deny

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma(4, 6, 8, 10, 12, 14)$

$$SOP = A F' + G F'$$

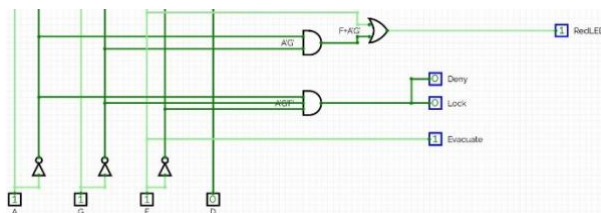


- RedLED :

1	1	1	1
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: $\Sigma(0, 1, 2, 3, 5, 7, 9, 11, 13, 15)$

$$SOP = F + A' G'$$

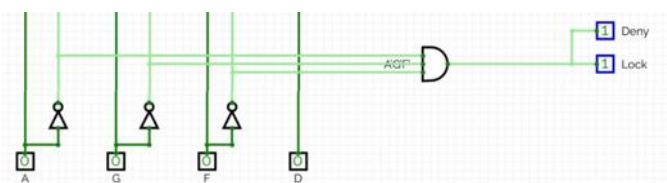


- Lock :

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

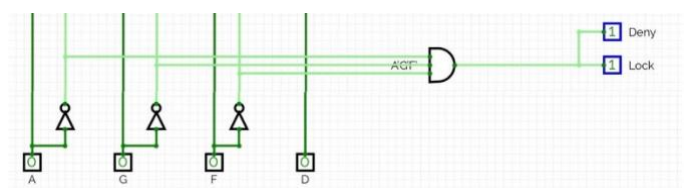
Minterms: $\Sigma(0, 2)$

$$SOP = A' G' F'$$



Minterms: $\Sigma(0, 2)$

$$SOP = A' G' F$$

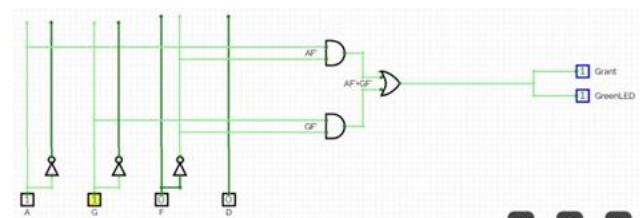


- GreenLED :

0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

Minterms: $\Sigma(4, 6, 8, 10, 12, 14)$

$$SOP = A F' + G F'$$

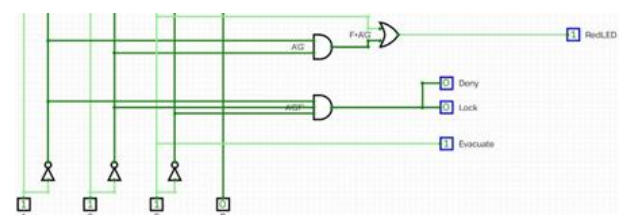


- Evacuate :

0	1	1	0
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: $\Sigma(1, 3, 5, 7, 9, 11, 13, 15)$

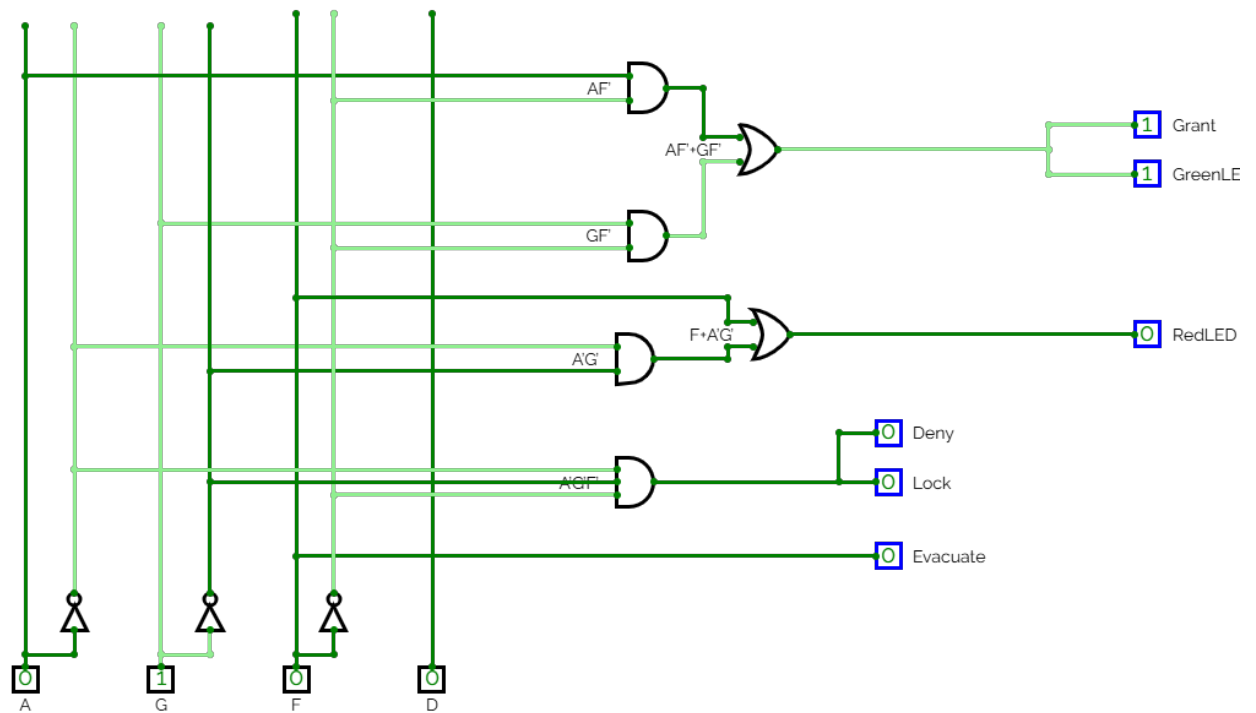
$$SOP = F$$



7. Circuit Implementation & Testing

The circuit was implemented in CircuitVerse using a structured vertical bus layout. This ensures clear organization and minimal gate usage.

https://circuitverse.org/simulator/embed/project_sen2



Scenario 3: Mini Vending Controller

1. Scenario Description

The Mini Vending Machine utilizes four binary inputs and six binary outputs.

2. Inputs:

- C1 (1-unit coin): High (1) if a 1-unit coin is inserted.
- C2 (2-unit coin): High (1) if a 2-unit coin is inserted.
- S (Select): High (1) if the Selection button is pressed.
- O (Out of stock): High (1) if the item is out of stock.

3. Outputs:

- Dispense: Activates the item dispensing mechanism.
- ReturnCoin: Returns inserted coins.
- ThankLED: Signals a “thank you” indicator.
- Change: Returns excess coin(s) as change.
- Error: Signals an invalid operation or insufficient funds.
- ServiceLED: Indicates service or fault condition.

3. System Priorities:

1. Out of Stock (O = 1): Dispense, ThankLED, Change = 0; Error, ServiceLED = 1; ReturnCoin = 1 if any coin inserted.

2. Normal Operation (O = 0):

- C1 = 1, C2 = 0, S = 1: Dispense item, ThankLED on, Change = 0.
- C2 = 1, C1 = 0, S = 1: Dispense item, ThankLED on, Change = 1.
- No coins (C1 = 0, C2 = 0, S = 1): Error = 1.
- Invalid coins (C1 = 1, C2 = 1): ReturnCoin = 1, Error = 1.

5. Truth Table Verification:

A 16-row truth table was constructed to verify all possible combinations of inputs.

C1	C2	S	O	Dispense	ReturnCoin	ThankLED	Change	Error	ServiceLED
0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	1	0
1	0	0	0	0	0	0	0	0	0
1	0	1	0	1	0	1	0	0	0
0	1	0	0	0	0	0	0	0	0
0	1	1	0	1	0	1	1	0	0
1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	1	0	0	0	0
0	0	0	1	0	0	0	0	0	1
0	0	1	1	0	0	0	0	0	1
1	0	0	1	0	1	0	0	0	1
1	0	1	1	0	1	0	0	0	1
0	1	0	1	0	1	0	0	0	1
0	1	1	1	0	1	0	0	0	1
1	1	0	1	0	1	0	0	0	1
1	1	1	1	0	1	0	0	0	1

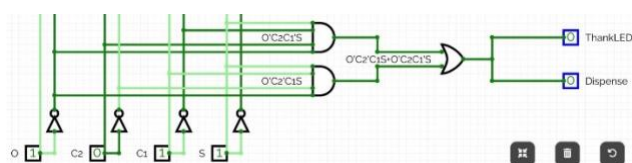
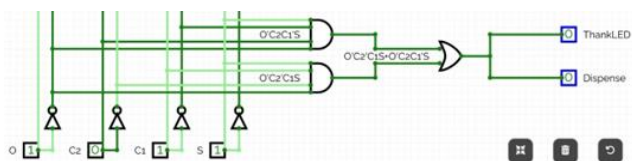
6. K-Map Simplification & SOP Equations

- Dispense :

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma(3,5)$

SOP= $O' C2' C1 S + O' C2 C1' S$

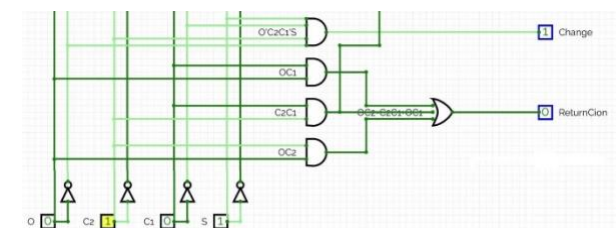
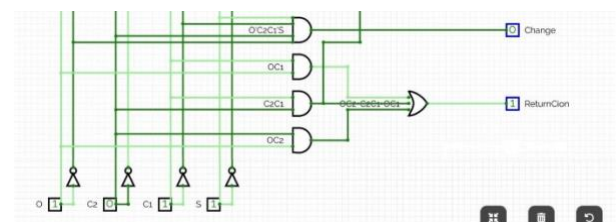


- ReturnCoin :

0	0	0	0
0	0	1	1
1	1	1	1
0	0	1	1

Minterms: $\Sigma(6,7,10,11,12,13,14,15)$

SOP = $O C2 + C2 C1 + O C2$

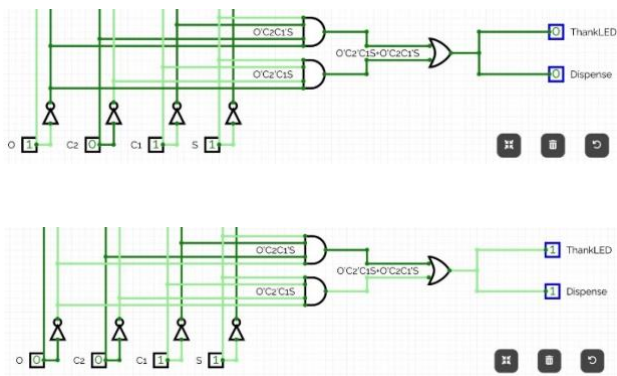


- ThankLED :

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (3,5)$

$$SOP = O' C2' C1 S + O' C2 C1' S$$

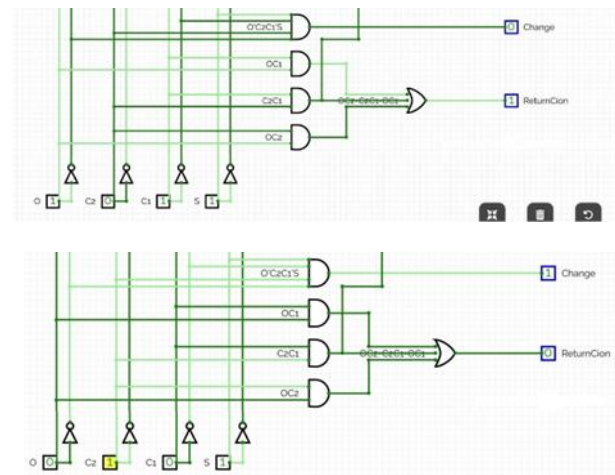


- Change :

0	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: $\Sigma (5)$

$$SOP = O' C2 C1'$$

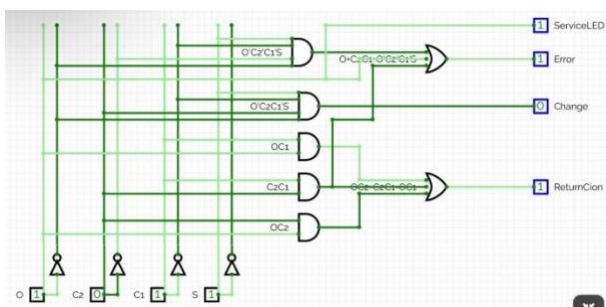


- Error :

0	1	1	0
0	1	1	1
1	1	1	1
1	1	1	1

Minterms: $\Sigma (1,6,7,8,9,10,11,12,13,14,15)$

$$SOP = O + C2 C1 + O' C2' C1' S$$

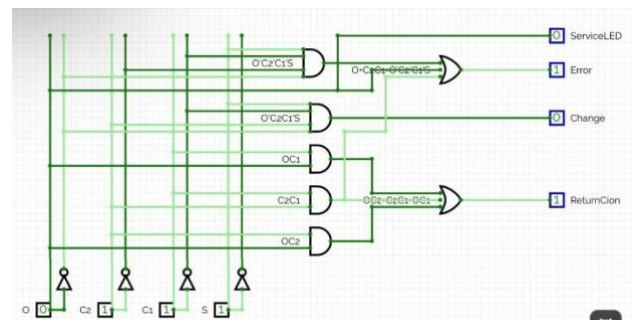


- ServiceLED :

0	0	0	0
0	1	0	0
1	1	1	1
1	1	1	1

Minterms: $\Sigma (8,9,10,11,12,13,14,15)$

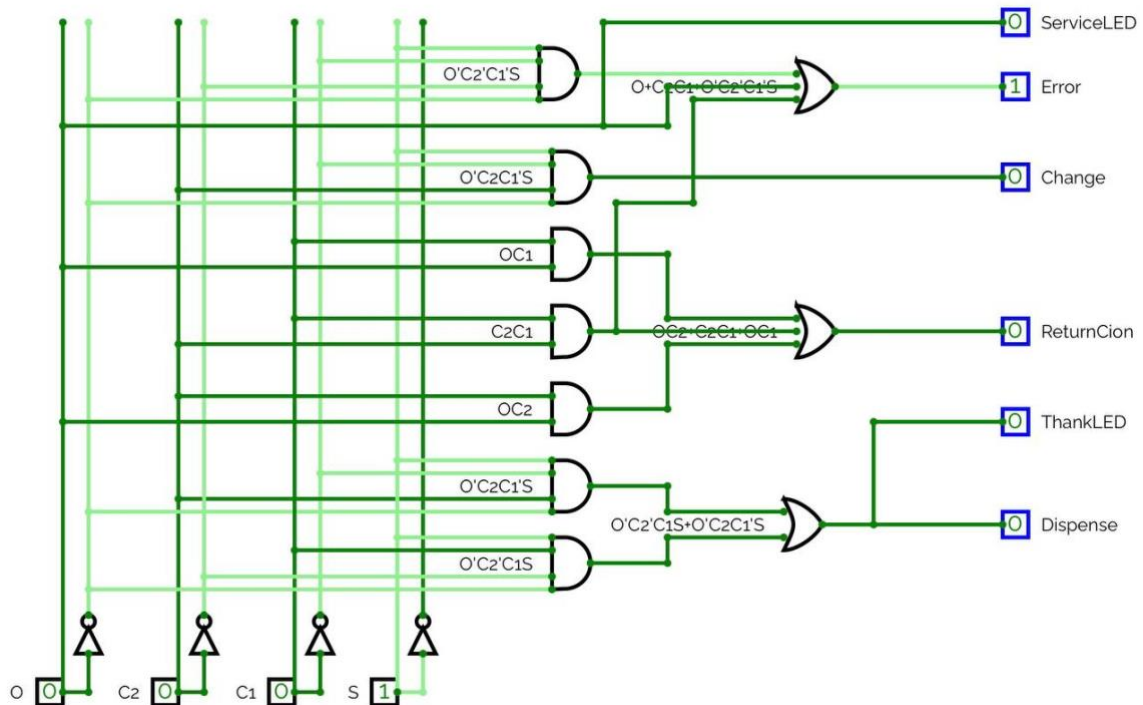
$$SOP = 0$$



7. Circuit Implementation & Testing

The circuit was implemented in CircuitVerse using a structured vertical bus layout. This ensures clear organization and minimal gate usage.

<https://circuitverse.org/simulator/embed/pro-d6ae11f7-1bb0-4a17-8521-ccc92b527ded>



8. Usability and Fault Handling

• Usability

The machine is simple and easy to operate : insert coins and press the selection button.
LED indicators guide the user:

- **ThankLED** → successful purchase
- **Error** → error or problem
- **Change** → change returned
- **ServiceLED** alerts the operator when the product is out of stock or there is a fault.

The system reduces errors and makes the process clear and smooth for all users.

• Fault Handling:

1. Inserting both coins simultaneously:

- If the user inserts both a 1-unit and a 2-unit coin at the same time, it is considered an invalid condition.

- The system returns all coins and lights the Error LED to alert the user.

2. Selecting the product when out of stock:

- If the user inserts money and presses the selection button but the machine is empty,
- The system returns the money, lights the Error LED, and activates the ServiceLED to alert the operator.

3. Inserting money and waiting without selection:

- If the user inserts coins but does not press the selection button, the machine remains idle.
- No product is dispensed, no LEDs light up, and no money is returned until the user presses the button or another condition occurs.

4. Selecting the product without inserting money:

- If the user presses the selection button without inserting any coins,
- The Error LED lights up and all other outputs remain 0 (no dispense, no ThankLED, no change).

Usage Suggestions

- **Scenario 2: Access Control & Indicators :**
University gate: Smart gate systems can replace security guards to control access to labs and buildings, with emergency priority (e.g., fire), enhancing safety and reducing reliance on human staff.
- **Scenario 3: Mini Vending Controller :**
Street machines: Can operate continuously without interruption, with a smart access system allowing smooth and secure public use, and clear status indicators to prevent operational issues.